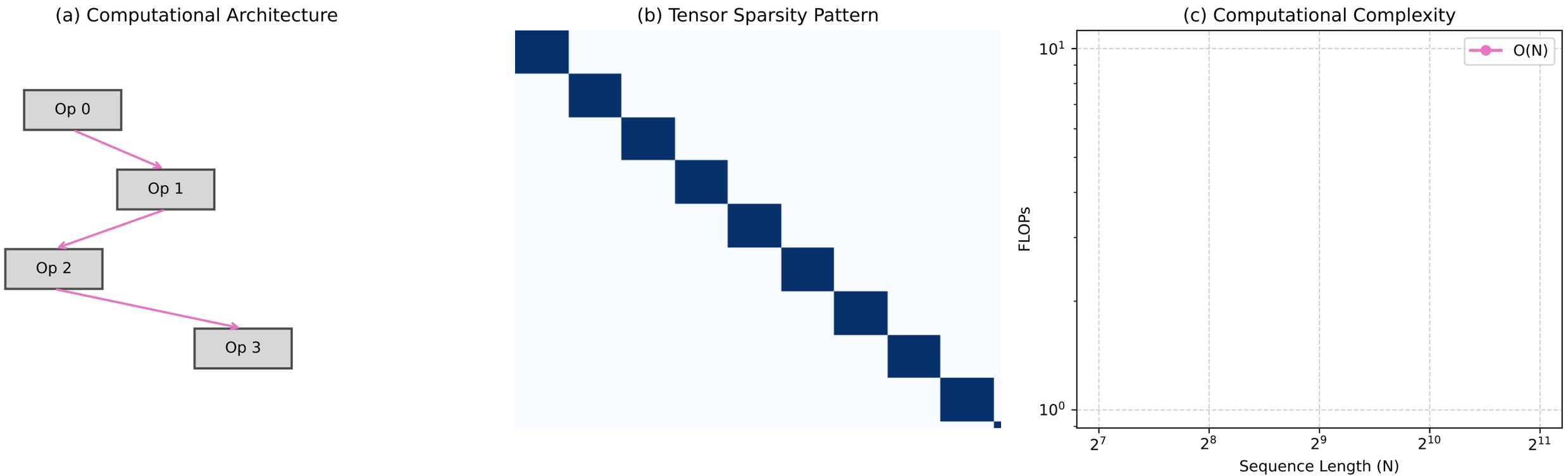Module Architecture: trainer_factory